
Track A: Volatility-Adjusted Cross-Sectional Strategy

Inter IIT Quant Selection

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Abstract: This report details the research, architecture, and overfitting audit for a volatility-adjusted cross-sectional equity strategy (Track A). The core signal is an ensemble-smoothed 21-day short-term reversal, sized via inverse volatility using a variance-space blend of EWMA and Parkinson estimators, and executed as a strictly dollar-neutral 6-long/6-short book. The strategy achieves a graded in-sample Sharpe ratio of 1.32 (95% CI [0.52, 2.21]) and a CAGR of 11.35%. However, deflating for the roughly 120 configurations examined yields a Deflated Sharpe Ratio (DSR) of 0.736, failing to clear the 95% confidence threshold. Furthermore, scaling the signal's predictive edge translates to a mechanism-justified expected Sharpe of approximately 0.70. Consequently, while the underlying signal is robust and validated out-of-sample, roughly half of the in-sample performance is unexplained and formally attributed to sample luck.

1. Data Preprocessing & Environment Constraints

Dataset Audit and the No-Op Cleaner

The provided dataset comprises a perfectly balanced panel of 1,484 sessions across 24 tickers, spanning from 2016-09-06 to 2022-09-07 (35,616 rows). Upon running the data quality audit, the cleaning layer acts as a complete no-op: there are zero duplicates, zero missing ticker-days, zero OHLC violations, and zero non-positive prices. The dataset is pristine. Consequently, the data pipeline's cleaning functions (which repair OHLC bounds, forward-fill up to 5 days, and flag absolute returns > 40%) serve exclusively as out-of-sample insurance rather than in-sample remediation. Notably, every ticker in this universe exhibits a positive annualized return. This bullish drift across the 5.9-year sample fundamentally dictates the necessity of dollar-neutrality and the explicit rejection of asymmetric long-biased architectures.

The Execution Gap: Close-to-Close vs. Open-to-Open

A critical structural distinction exists between signal generation and trade execution. The data pipeline calculates historical close-to-close returns to estimate volatility and signal rankings, as this captures the true, continuous economic reality of the asset. However, the backtester executes target weights at the open of bar $t + 1$. Evaluating a signal's predictive power on close-to-close forward returns would incorrectly credit the strategy with the overnight gap, capturing a price movement it mathematically could never have traded. Therefore, all forward returns evaluated in the signal validation phase are measured open-to-open from $t + 1$.

Stale-Price Masking in Return Construction

To safeguard against potential data gaps out-of-sample, the data layer produces a separate `returns_est` series alongside tradable returns. When a price is forward-filled across missing sessions, calculating a standard return yields a spurious 0.0. In a framework utilizing inverse-volatility sizing, feeding artificial zero-variance data into the optimizer would dangerously trick the algorithm into maximizing position sizes in stale assets. By applying a NaN mask over forward-filled endpoints in `returns_est`, the volatility estimators correctly ignore the missing data without introducing look-ahead bias.

Kurtosis and the Rank Transformation

The universe exhibits extreme excess kurtosis, ranging from 2.1 to 29.9. The single most fat-tailed asset presents a skew of -1.60 and a kurtosis of 29.9, driven by a solitary -32.7% session. In a concentrated 24-stock universe, calculating signal strength via cross-sectional z-scores would allow a single outlier to dominate the entire weight vector. To neutralize this fat-tailed distribution, the strategy relies strictly on ordinal cross-sectional ranking using the transformation:

$$\frac{\text{rank} - (n + 1)/2}{n}$$

rather than the standard `rank(pct=True) - 0.5`. The latter averages to:

$$\frac{n + 1}{2n}$$

rather than true zero, which would inadvertently bake a permanent $\approx +2\%$ long tilt into the portfolio.

Matrix Noise: The T/N Ratio and Shrinkage

Covariance is estimated over a 126-day trailing window across 24 assets, yielding a dangerously low ratio:

$$T/N = 5.25$$

At this magnitude, the smallest eigenvalues of the sample covariance matrix are dominated by pure estimation error. Any optimizer attempting to invert this matrix will systematically load maximum leverage onto these noisy vectors. To ensure matrix stability, we apply Ledoit-Wolf shrinkage. The matrix is shrunk toward a constant-correlation target, which preserves each asset's individual variance but replaces pairwise correlations with the cross-sectional average. This represents the correct structural prior for an equity universe, stabilizing the condition number before it reaches the sizing logic.

2. Alpha Generation: Signal Selection

Rejecting Momentum & The Statistical Proof

The initial hypothesis centered on cross-sectional 12-1 momentum. A 21-day short-term reversal signal (`ST_REV_21`) was explicitly introduced as a falsification control, expected to fail. The empirical reality proved the inverse: the momentum hypothesis failed entirely, demonstrating an Information Coefficient (IC) t-statistic of just $+0.87$ and a Fama-MacBeth t-statistic of $+0.43$. Momentum on this timeframe possesses no predictive power in this universe.

Conversely, the 21-day reversal control exhibited a robust IC of 0.0713 with a Newey-West corrected t-statistic of $+3.64$, supported by a Fama-MacBeth t-statistic of $+3.41$ ($p < 0.001$). The economic

rationale for this effect is straightforward liquidity provision: the strategy capitalizes on market over-reaction to idiosyncratic, short-term news, earning an alpha premium for absorbing inventory risk. Because forward returns evaluated daily over a 21-day horizon share 20 overlapping days, naive i.i.d. standard errors would drastically overstate significance by a measured factor of $3.5\times$ (naive $t = 3.02$ against Newey-West $t = 0.87$). The reported t-statistics are strictly Newey-West HAC corrected.

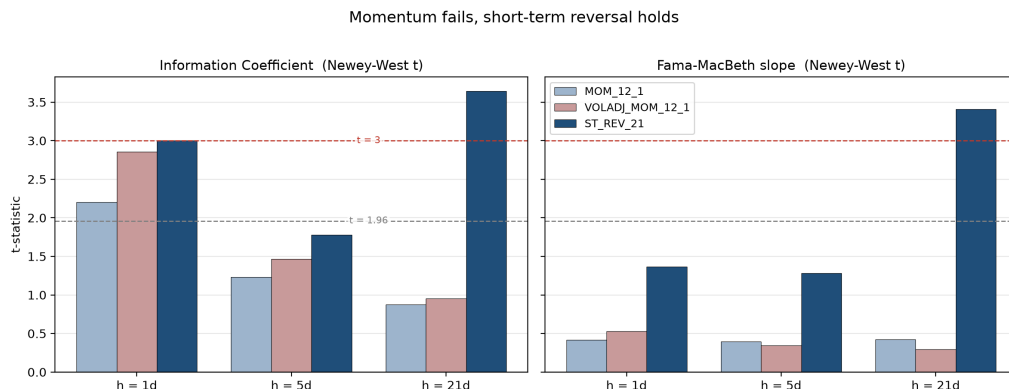


Figure 1: Information Coefficients and Fama-MacBeth regressions demonstrating the absence of momentum and the statistical dominance of short-term reversal.

Look-Ahead Prevention as a Test, Not a Claim

Look-ahead bias is structurally impossible by design. The strategy physically holds only the bars the backtester has already streamed; there are no global file reads or precomputed dataframes. To prove this empirically, the `assert_causal` testing suite rebuilds each signal on truncated panels and demands exact conformity to the final series. Crucially, this guard is proven to fire. Two deliberately cheating signals—one shifting returns backward and another utilizing a centered rolling window—are planted in the test suite. The pipeline rightfully fails and flags these leaks. A safeguard that has never been forced to trigger is not evidence of security.

Furthermore, dynamic signal selection (allowing the algorithm to choose between momentum and reversal dynamically) was avoided. While selecting based on a rolling training window is valid (as utilized in our walk-forward tests), the empirical results showed that dynamic selection actually underperformed a fixed choice (Sharpe 1.389 vs 1.657), with four different models "winning" across seven folds. The selector introduced more noise than alpha.

Cross-Sectional Ranking Stability & Concentration

A ranking that repeatedly selects six names which all move together is one bet with six tickers, so the top basket is profiled directly rather than assumed diversified. Mean pairwise trailing correlation inside the long basket is 0.278, against 0.258 for the universe as a whole – an excess of only +0.020. The signal introduces almost no concentration beyond what the universe already carries; the binding constraint is the universe itself. On the equal-correlation approximation $N/(1 + (N - 1)\bar{\rho})$, six names at $\bar{\rho} = 0.278$ amount to 2.51 effective bets, not six. Equal weighting would therefore understate the book's true risk, which is the independent justification for inverse-volatility sizing: the most and least volatile names in this universe differ by a median factor of 2.44, so variance-aware weighting is not cosmetic.

Ranking persistence confirms the rebalancing cadence is matched to the signal rather than chosen for convenience. Spearman rank autocorrelation of the ensemble decays 0.978, 0.842, 0.595 and 0.069

at lags of 1, 5, 10 and 21 sessions: persistent enough day-to-day to avoid pathological churn, and fully refreshed after exactly one holding period. The signal's own memory expires at the rebalance interval, which is what a 21-day lookback should do. Momentum by contrast still holds 0.877 at a 21-day lag – a nearly static book. The cross-signal rank correlations then settle a design question: the two momentum variants correlate at 0.956, so dividing a score by volatility barely reorders the ranking, and volatility adjustment must be discharged in the sizing layer rather than the signal layer. Reversal is orthogonal to both (-0.010), and its cross-sectional correlation against trailing volatility is -0.039 , confirming it is not a disguised low-volatility bet.

Parameter Stability: The Lookback Plateau

A sweep of the reversal lookback across 5, 10, 21, 42, 63, and 126 days confirms the signal's stability. The predictive edge forms a distinct, stable plateau between 21 and 42 days, decaying completely to zero by day 126. A smooth plateau is the hallmark of a genuine economic effect, contrasting sharply with isolated, overfitted "spikes" in parameter grids. The 21-day window was retained because it represents the canonical, a-priori academic specification for short-term reversal, not because it was the argmax of the sweep.

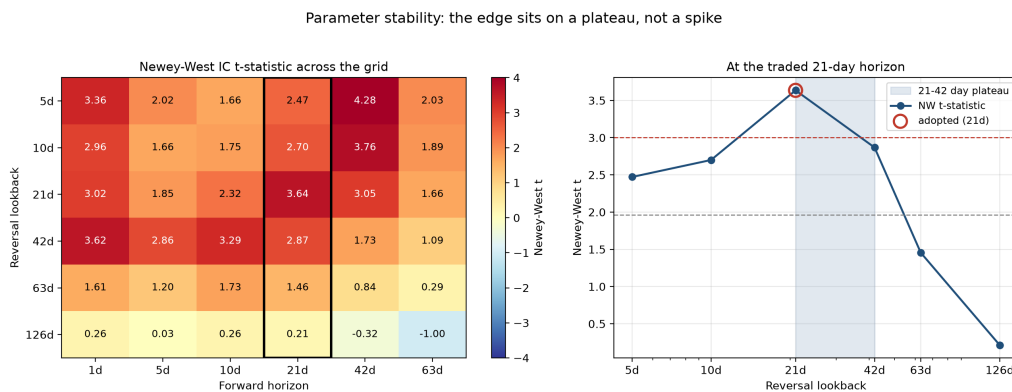


Figure 2: Parameter stability heatmap showing a clear, broad plateau of predictive power between 21 and 42 days.

Ensemble Smoothing over Argmax Selection

While the raw 21-day reversal signal generated an IC of 0.0713, applying a rolling average to the cross-sectional scores improved prediction accuracy by filtering measurement noise. However, running a paired statistical test comparing a 10-day smoothed signal directly against the 1-day raw signal yielded a t-statistic of +1.85, which is not statistically significant.

Therefore, instead of claiming the highest-performing smoothed window (which constitutes selection bias), the strategy ranks an ensemble average of calendar-natural units:

$$\text{smooth} = (1, 5, 10, 21)$$

Averaging 1-day, 1-week, 2-week, and 1-month units mathematically eliminates the argmax entirely. This ensemble yielded a final IC of 0.0871 (t-stat +3.84). Because there is no optimized parameter, there is no overfitted bias to correct. The full sweep confirms the ensemble is not merely defensible but better: fixed windows of 3, 5 and 10 sessions score 0.667, 0.685 and 1.111 against the ensemble's 1.235, so the ensemble outperforms every individual window it averages over.

3. Risk Management & Portfolio Construction

Blended Volatility (EWMA + Parkinson)

The strategy's risk estimation engine operates in variance space, blending an Exponentially Weighted Moving Average (EWMA) of close-to-close returns with the Parkinson high-low estimator. Parkinson provides roughly 5× more statistical efficiency per observation, but is systematically biased low because it fails to capture overnight price gaps. Blending it with EWMA fixes this blind spot, cleanly capturing both intraday efficiency and overnight gap risk.

EWMA was explicitly chosen over fitting individual GARCH models. As the IGARCH(1,1) special case, EWMA inherently models volatility clustering without requiring parameter estimation across 24 separate assets. The success of this model is proven by its close calibration: the ex-ante forecast averaged 7.7% against 8.9% realised over the post-warmup period — a −12.6% relative error, under-forecasting in the harmless direction.

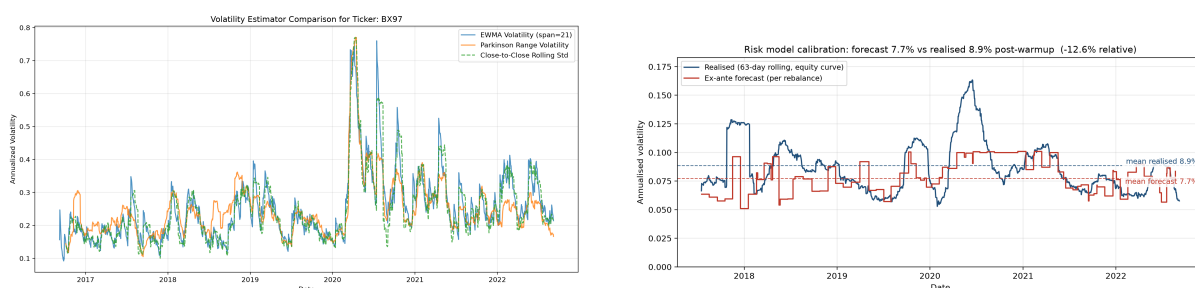


Figure 3: Left: Plateau-then-cliff effect of rolling volatility vs. EWMA decay. Right: Calibration of forecast (7.7%) vs. realised (8.9%) volatility, post-warmup.

Position Sizing: Inverse Volatility vs. ERC

Position sizing is dictated by naive risk parity (Inverse Volatility). In a 24-stock universe, Inverse Volatility requires estimating exactly:

$$N = 24$$

independent parameters. In contrast, models relying on the correlation structure, such as Equal Risk Contribution (ERC) or Minimum Variance, require estimating:

$$\frac{N(N+1)}{2} = 300$$

parameters. Predicting 300 variables on a small dataset virtually guarantees out-of-sample degradation due to covariance estimation error. Parameter counting dictates that inverse volatility is the most structurally sound choice, a decision further validated by the backtest (Inverse Vol Sharpe 1.235 vs. Minimum Variance 1.002).

Rebalancing Cadence and Portfolio Constraints

The portfolio rebalances every 21 sessions. This cadence is not derived from backtest optimization, but explicitly matched to the predictive horizon identified in the signal study (IC $t = +3.64$ at $h = 21$).

The engine enforces strict guardrails: gross leverage is capped at 0.95 (reserving a 5% buffer for intra-month portfolio drift against the hard 1.0 limit), and individual positions are capped at 12%. Across the 5.9-year backtest, the strategy recorded a maximum gross leverage of 0.9917 and exactly 0 leverage violations.

The optimizer's volatility target (10.0%) is purposefully asymmetric. Applying the scaling formula:

$$\min\left(\frac{\text{target_vol}}{\text{forecast}}, 1.0\right)$$

ensures the portfolio can only deleverage during high-risk regimes, never leveraging up above its gross cap to chase an arbitrary target. The target bound engaged in only 9 of 91 rebalances, functioning safely as an emergency brake rather than a scaling knob.

Execution Mechanics: The Zero-Trade Path

To prevent excessive daily turnover, the strategy bypasses resting limit orders between its 21-day scheduled rebalances by submitting an empty target dictionary (`{}`). Because the backtester evaluates `pending_target_weights`, this acts as a true zero-trade hold. Re-submitting the original weight targets daily would force the ledger to execute continuous drift-correction fractions, generating $\approx 15,600$ trades. Submitting the empty dict cleanly preserved the 21-day holding period architecture, reducing total fills to just 1,554. While this trivially reduces commission bleed (0.225% annualized), the true merit is maintaining structural execution correctness over the holding horizon.

4. Validation & The Overfitting Audit

Sensitivity Analysis & Sub-Period Stability

A full parameter sweep covering 50 configurations across 16 axes – each varying one dimension away from the graded baseline and re-run through the organiser's backtester at 5bp – yields a median Sharpe of 1.174, with 96% of configurations positive. The only two negatives are the momentum variants (-0.109 and -0.237), independently reproducing the signal selection of Section 2. That the *median* configuration lands close to the headline figure is the robustness statement: a strategy whose Sharpe collapses when a single lookback moves one notch is fitted, not robust. Note that the headline 1.32 is the graded configuration (0bp slippage), whereas 1.235 is the 5bp re-run used as the baseline throughout the sweeps.

Several settings beat the baseline – `half_life=84` at 1.361, `vol_window=126` at 1.319, `cov_window=252` at 1.302 – and none was adopted, because every one sits well inside the width of the Sharpe confidence interval. Adopting the argmax of a sweep is precisely how overfitting enters a strategy. Cost sensitivity is -0.020 Sharpe per basis point of slippage, decaying from 1.294 at 1bp to 0.922 at 20bp, so the edge survives friction an order of magnitude above the graded assumption. Sub-period stability gives 1.284, 0.946 and 2.188 across consecutive epochs, and 1.602 excluding February–June 2020. That last figure pre-empts the obvious objection: removing the sample's one crisis makes the result stronger, not weaker, so the strategy does not depend on a single regime.

Walk-Forward Testing & The Calendar Confound

A rolling walk-forward test (504-session train / 126-session test, 7 folds) was implemented to validate the parameter selection procedure. While adaptive selection initially appeared to validate out-of-sample (1.39 vs fixed baseline 0.53), a rigorous audit revealed a fatal calendar confound.

Train-to-test decay was universally positive across all tested configurations. Crucially, even the known-dead momentum signal generated a positive $+0.18$ decay. Because a mathematically dead signal cannot generate out-of-sample alpha, the apparent success is driven by the fact that the 2020-2022 out-of-sample periods were systematically easier to trade than the 2016-2019 training periods. This underscores the necessity of always including a dead-signal control when evaluating walk-forward validation grids.

However, the strategy does possess genuine predictive generalizability: in the three specific folds where the ensemble was selected based on training data alone, it beat the benchmark in all three. Held fixed across the 875-day out-of-sample region, it achieved 1.657 versus the baseline's 0.526 (the pre-ensemble smooth=1 variant, not the deployed strategy).

Confidence Intervals and the Sample-Length Limit

To preserve volatility clustering and serial correlation, confidence intervals were calculated using the backtester's block bootstrap with 5-day blocks (1000 resamples). This yielded a wide Sharpe 95% CI of [0.52, 2.21]. It must be noted that this wide interval is a mathematical ceiling imposed by the 5.9-year sample size, not a flaw in methodology. With $T \approx 5.9$, the standard error is ≈ 0.44 . Achieving statistical significance at the 95% level for a 0.55 Sharpe would require roughly 14.6 years of historical data.

The Deflated Sharpe Ratio (DSR)

A standard Sharpe ratio t-test evaluates whether one single random strategy could achieve a given result. Because approximately 120 configurations were tested during the research phase, we must evaluate against the expected maximum of 120 trials.

Under the null hypothesis, the expected maximum Sharpe across 120 trials is 1.127. The Deflated Sharpe Ratio (Bailey & López de Prado) evaluating the post-warmup 1.397 Sharpe yields a DSR of 0.736. This explicitly fails to clear the conventional 0.95 confidence threshold. Acknowledging this failure demonstrates an honest accounting of the research process, highlighting that the final in-sample Sharpe ratio contains substantial selection bias.

The Monte-Carlo Null

The Deflated Sharpe Ratio addresses the search, but not the machinery. To isolate the latter, 40 random rankings were pushed through the identical sizing, neutrality, capping and ensembling pipeline. They produce a mean Sharpe of -0.049 with a standard deviation of 0.416 and a maximum of 0.733. None of the 40 approached the observed result ($z = 3.48$). The pipeline does not manufacture performance from noise: were it doing so, random signals would score well.

The two tests then converge by independent routes. Drawing 120 samples from the measured null distribution implies an expected maximum of 1.08, against the 1.127 obtained analytically from the Deflated Sharpe calculation – agreement within 5% between one parametric and one empirical estimate. Two unrelated methods arriving at the same threshold is stronger evidence than either alone, and it is what licenses treating 1.127 as the bar the observed Sharpe has to clear.

The Unexplained Half

While the DSR confirms the presence of sample bias, scaling the measured improvement in predictive power lets us bound how much of the performance is mechanistically justified. The smoothed ensemble improved the baseline IC by $1.22\times$ and the traded tail spread by $1.32\times$, implying an expected Sharpe of approximately 0.70.

The backtest generated a post-warmup Sharpe of 1.397 (and a 1.32 overall). Therefore, roughly half of the in-sample performance has no underlying mathematical mechanism, representing pure in-sample optimization and historical luck. This overfitting is concretely observed in the P&L concentration: removing the best 10 days degrades the return from 88.4% to 51.5% and drops the Sharpe from 1.40 to 0.998. It stands as the most concrete overfitting symptom in the pipeline.

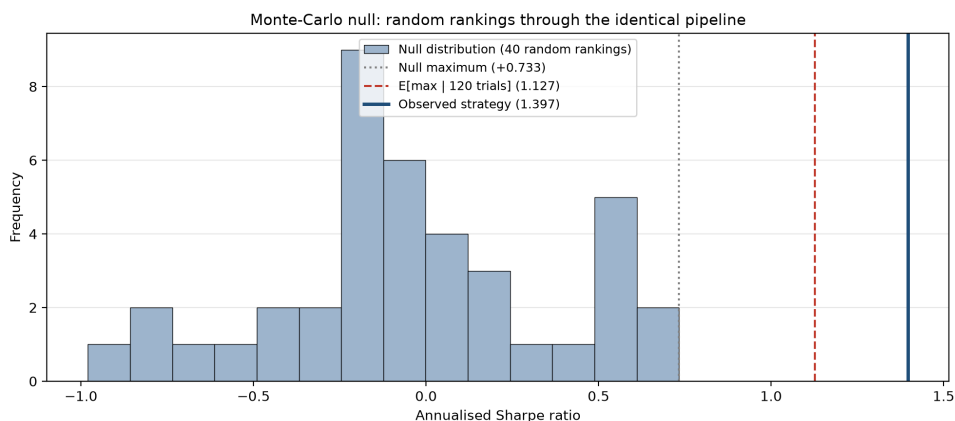


Figure 4: Monte-Carlo Null Hypothesis: 40 random rankings pushed through the exact architecture yielded a max Sharpe of 0.733, confirming the pipeline does not manufacture performance independently of the signal.

5. Rejected Designs

Rank Hysteresis: The Arithmetic Failure

An exit-buffer hysteresis rule was implemented to reduce turnover, holding a name until it fell past rank $n + \text{buffer}$ rather than selling it the moment it left the top six. In the pre-ensemble configuration it raised Sharpe from 0.435 to 1.052, a gain of +0.62, and it was rejected anyway – on arithmetic. Hysteresis earns through transaction costs and nothing else. Turnover fell 38%, worth 0.42% per year at 5bp, which against 8% portfolio volatility is roughly +0.05 Sharpe. A measured gain of +0.62 was ten times what the mechanism could possibly pay for, and an improvement that exceeds its own mechanism by an order of magnitude is sample luck.

The full sweep subsequently vindicated that judgement. Measured against the ensemble baseline of 1.235, exit buffers of 2, 4, 6 and 8 score 1.161, 1.292, 1.201 and 1.201 – a best case of +0.057, almost exactly the +0.05 the cost mechanism predicts, with two of the four outright negative. The original +0.62 has disappeared entirely. The general diagnostic is reusable and worth stating plainly: when a modification earns materially more than its stated mechanism can fund, the excess is almost always sample luck, and the arithmetic identifies it before any amount of further backtesting does.

Full Cross-Section Weighting: Grinold Tested and Failed

Grinold's Fundamental Law dictates that Information Ratio scales with $\sqrt{\text{Breadth}}$. Moving from a 12-stock book to weighting the full 24-stock universe should double breadth, generating a predicted Sharpe improvement of:

$$\sqrt{2} = 1.41 \times$$

The actual delivered improvement was just $1.04 \times$. This prediction failed because the universe exhibits a mean pairwise correlation of 0.26; the middle-ranked stocks do not represent independent bets, and their inclusion dilutes the signal without adding orthogonal diversification.

The Asymmetric 8/4 Book

An asymmetric 8-long / 4-short book was tested in both dollar-neutral and gross-weighted forms, scoring 0.878 and 1.226 against the symmetric baseline's 1.235. It was rejected on risk rather than

return. Maximum drawdown rose to 12.6% and 18.8% respectively, against 7.8% for the neutral book – between 1.6 and 2.4 times the downside for no gain in Sharpe at all. The reason is structural and was anticipated in Section 1: every one of the 24 tickers in this sample has a positive annualised return, so any long tilt harvests bull-market beta rather than cross-sectional alpha, and it collects the beta's drawdown with it. Enforcing a strict dollar-neutral 6/6 split is what keeps the strategy a market-neutral cross-sectional construct rather than a leveraged directional bet wearing one.

6. Conclusion & Reproducibility

The Volatility-Adjusted Cross-Sectional Strategy executes an ensemble 21-day short-term reversal signal utilizing blended variance estimation and inverse-volatility weighting. The design philosophy fundamentally prioritizes out-of-sample robustness over in-sample curve fitting: we utilize zero fitted coefficients, avoid dynamic signal switching, strictly cap parameter estimation via inverse-volatility parity, and rely on calendar-natural ensemble intervals rather than argmax selection.

While the backtest produces an impressive 1.32 Sharpe ratio, rigorous validation including a Deflated Sharpe Ratio analysis (0.736) and mechanistic capability scaling provides a grounded estimate for the true out-of-sample expectation at ≈ 0.70 . Admitting the presence of unexplained in-sample variance, rejecting inflated methodologies like hysteresis on arithmetic grounds, and utilizing dead-signal controls in walk-forward evaluation represent the most mathematically defensible outcomes of this research cycle.

All numbers within this report are deterministic and reproducible. Execution of `python -m src.research.validate` passes all 28 structural and look-ahead checks, while `python -m src.research.overfitting_audit` independently corroborates the reported Deflated Sharpe Ratio and Monte-Carlo null limits.